

● EASY

● ALGEBRA

● ANSWER KEY

Systems of two linear equations in two variables

04

6 answers · Rationale-rich · Teach from doc

Q1**A**

A. $c = 4m$
 $c + m = 25$

Choice A is correct. It's given that Connor has c dollars, Maria has m dollars, and Connor has 4 times as many dollars as Maria. This can be represented by the equation $c = 4m$. It's also given that together, Connor and Maria have a total of \$ 25.00, which can be represented by the equation $c + m = 25$. Therefore, the system consisting of the equations $c = 4m$ and $c + m = 25$ represents this situation.

Choice B is incorrect. The equation $m = 4c$ represents a situation where Maria has 4 times as many dollars as Connor, rather than the situation where Connor has 4 times as many dollars as Maria.

Choice C is incorrect. The equation $c = 25m$ represents a situation where Connor has 25 times, rather than 4 times, as many dollars as Maria. The equation $c + m = 4$ represents a situation where Connor and Maria together have a total of \$ 4.00, rather than \$ 25.00.

Choice D is incorrect. The equation $m = 25c$ represents a situation where Maria has 25 times as many dollars as Connor, rather than the situation where Connor has 4 times as many dollars as Maria. The equation $c + m = 4$ represents a situation where Connor and Maria together have a total of \$ 4.00, rather than \$ 25.00.

Q2**B**

B. $26b + 35g = 881$
 $b + g = 28$

Choice B is correct. Outfits for boys cost \$26 each and the teacher ordered b outfits for boys, so the teacher spent $26b$ dollars on outfits for boys. Similarly, outfits for girls cost \$35 each and the teacher ordered g outfits for girls, so the teacher spent $35g$ dollars on outfits for girls. Since the teacher spent a total of \$881 on outfits for boys and girls, the equation $26b + 35g = 881$ must be true. And since the teacher ordered a total of 28 outfits, the equation $b + g = 28$ must also be true.

Choice A is incorrect and may result from switching the constraint on the total number of outfits with the constraint on the cost of the outfits. Choice C is incorrect and may result from switching the constraint on the total number of outfits with the constraint on the cost of the outfits, as well as switching the cost of the outfits for boys with the cost of the outfits for girls. Choice D is incorrect and may result from switching the cost of the outfits for boys with the cost of the outfits for girls.

Q3**A**

A. 10

Choice A is correct. According to the first equation in the given system, $y = 4$. Substituting 4 for y in the second equation in the given system yields $x = 4 + 6$, or $x = 10$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of y , not x .

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B****B.** 2, 2

Choice B is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 2$, or at the point 2, 2. Therefore, the solution x, y to the system is 2, 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**B****B.** 5, -3

Choice B is correct. A solution to a system of equations in the xy -plane is a point x, y that lies on the graph of each equation in the system. The first equation given is $x = 5$. Substituting 5 for x in the second given equation yields $y = 5 - 8$, or $y = -3$. It follows that in the xy -plane, the point 5, -3 lies on the graph of each equation in the system. Therefore, the solution to the given system of equations in the xy -plane is 5, -3.

Choice A is incorrect. The point 0, 0 doesn't lie on the graph of either equation in the given system.

Choice C is incorrect. The point 5, -8 doesn't lie on the graph of the second equation in the given system.

Choice D is incorrect. The point 5, 8 doesn't lie on the graph of the second equation in the given system.

Q6

B

B. 6

Choice B is correct. Adding the second equation in the given system to the first equation in the given system yields $3x + (-3x + y) = 12 + -6$, which is equivalent to $y = 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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